

OBJECT DETECTION Using Lidar Module

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Radio detection and ranging (radar) systems detect the presence and distance of objects using radio frequencies. The

distance of an obstacle is calculated from the time taken for the radio waves to travel to the object and, after reflection from it, return back.

Light detection and ranging (lidar) systems work much the same way; the only difference is they use laser light instead of radio waves.

This simple and affordable project uses the concept of finding an object using a lidar module. The VL53L0X lidar sensor module used here is available as a breakout board.

The lidar module sends laser pulses and calculates the distance of an object from it using the time of flight. It oscillates to find the obstacle within its range and displays the same on a graphic liquid crystal display (GLCD).

Circuit and description

The circuit diagram for object detection using lidar is shown in Fig. 1.

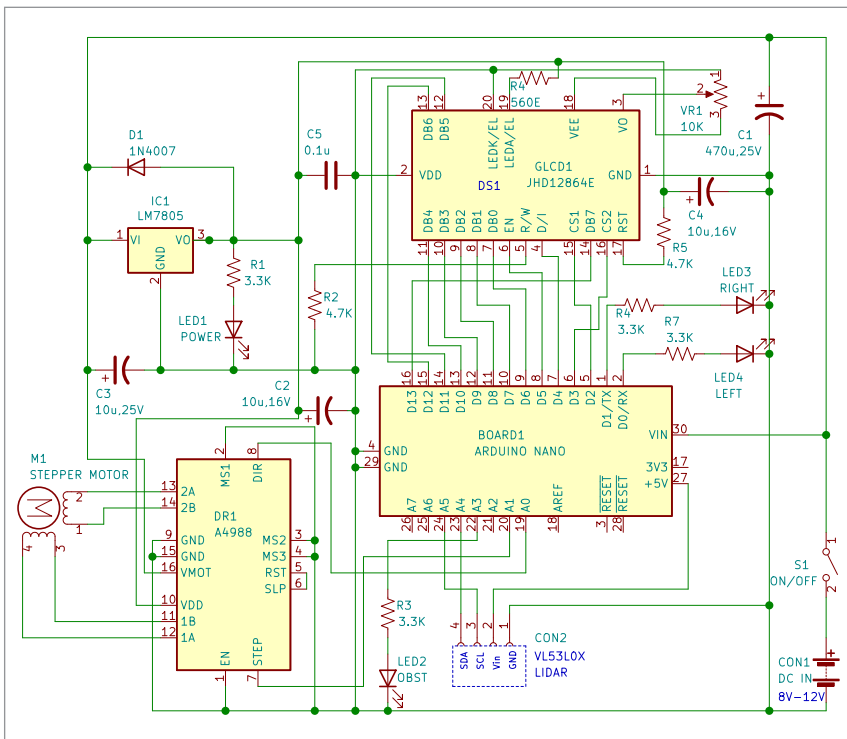


Fig. 1: Circuit diagram



Fig. 2: Lidar module attached to stepper motor



Fig. 3: Author's prototype with screen showing distance of the object and the scanning angle

PARTS LIST

Semiconductors:

- IC1 - 5V voltage regulator, LM7805
- BOARD1 - Arduino Nano board
- DR1 - A4988 stepper motor driver module
- LIDAR - VL53L0X LiDAR sensor module
- D1 - 1N4007 rectifier diode
- LED1-LED4 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3, R6, R7 - 3.3-kilo-ohm
- R2, R5 - 4.7-kilo-ohm
- R4 - 560-ohm
- VR1 - 10-kilo-ohm pot/preset

Capacitors:

- C1 - 470µF, 25V electrolytic
- C2-C4 - 10µF, 16V electrolytic
- C5 - 0.1µF ceramic

Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2 - 4-pin female Berg connector
- GLCD1 - 128×64 JHD12864E GLCD module
- M1 - 4-wire stepper motor
- S1 - SPST on/off power switch
- 8 to 12V DC power supply
- Heat-sink for IC1